

The Mathematics of the pmsimstats-ng Manuscript Compendium

A unified account of the notation, data-generating processes, and analysis models across the eleven manuscripts under `analysis/report/`

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Abstract

The eleven manuscripts of the pmsimstats-ng compendium share one outcome model, one family of trial designs, and one analysis-model convention, but each paper varies a different piece of that shared machinery to ask a different question. This document collects, in one place and in the compendium’s own notation, the mathematics that each manuscript actually states: the response trajectory, the treatment- exposure and carryover mechanics, the two (and, jointly, three) architectures by which a biomarker moderates treatment response, a latent-variable reading of the covariance-moderation architecture, the decomposition of the treatment effect into its biological, placebo, and natural-history parts, and the inferential machinery used to test and calibrate the resulting interaction estimand. Where a manuscript’s own presentation is thin, inconsistent, or silent on a point the compendium’s design nonetheless raises, that gap is flagged explicitly rather than papered over with new derivation. Every equation below is attributed to the manuscript it came from.

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1 Scope and reading order

The compendium’s eleven manuscripts fall into two families. Nine share the biomarker-treatment interaction $\beta_{bm:D}$ as their estimand and the common outcome components of Section 2.2; one (Paper 04) targets the treatment *main* effect β_D under a simpler, biomarker-free model; one (Paper 10) is written in deliberately general, non-N-of-1-specific notation to isolate what drives realized Type I error in *any* Wald test on a mixed-model interaction coefficient. This document orders the material so that each section can be read against the one before it:

- 1. Notation** (§2) fixes indices, the sign convention, and the rule that N is always a total across randomization paths.
- 2. The response trajectory** (§3, Papers 06–07) gives the modified-Gompertz curve that every latent factor in the outcome follows, and the alternative curve families used to test whether the compendium’s conclusions depend on that choice.

3. **Trial designs, exposure, and carryover** (§4, Papers 02, 05) gives the four designs, the treatment-exposure indicators, and the carryover decay families that every later section takes as fixed infrastructure.
4. **Two architectures for a biomarker-treatment interaction** (§5, Paper 01) is the compendium’s central methodological contribution: the same interaction can live in the outcome’s first moment (mean moderation) or its second moment (covariance moderation), and the two are not the same model wearing different clothes.
5. **A latent-variable reading of covariance moderation** (§6, Paper 03) asks what generative mechanism a fitted covariance-moderation model is secretly standing in for, and gives the moment conditions under which a two-class mixture and a single MVN are observationally close.
6. **The dual-channel architecture** (§7, Paper 11) activates both channels at once and shows their contributions to the interaction test are additive to first order, with the two channel weights not separately identifiable from a single fitted coefficient.
7. **Partitioning the treatment effect** (§8, Paper 06) decomposes the outcome itself into biological, placebo-belief, and natural-history components, and derives closed-form bias identities for what a one-component analysis actually estimates when the true generative process has three.
8. **Informative dropout** (§9, Paper 09) gives the hazard mechanism used to censor participants as a function of their own symptom trajectory, and the missingness classification that follows from it.
9. **Inference: from a fitted coefficient to a calibrated test** (§10, Papers 08, 10, 02) collects the analysis-model machinery: a closed-form repeated-measures ANOVA derivation, a general four-channel decomposition of realized Type I error, and the cluster-robust sandwich-variance family used to correct it.
10. **The treatment main-effect track** (§11, Paper 04) is the compendium’s odd paper out: no biomarker channel at all, a linear (not Gompertz) DGP, and the classical closed-form power theory of Senn and of Zucker et al. that the rest of the compendium replaces with simulation.
11. **Shared performance measures** (§12) collects the Monte Carlo standard error, power, and calibration-ratio definitions used identically across every paper.

Every symbol is defined at first use and, where the compendium’s own `NOTATION.md` governs it, cross-referenced to that governing rule. Sign convention throughout: the three outcome components are non-negative *reductions* in symptom severity, so treatment and interaction coefficients are negative and moderation parameters are positive (`NOTATION.md` Rule 5).

2 Notation and conventions

2.1 Indices and structural counts

Symbol	Meaning
i	participant index
t	timepoint index (weekly measurement occasions throughout)
j, ij	period or occasion index within participant (Papers 04, 06, 08)
N	total number of participants, across all randomization paths
P	number of randomization paths in a design; expected allocation N/P per path
$n_{\text{reps}}, n_{\text{sim}}$	Monte Carlo replicates per cell
k	cycles per participant in a design sweep; also the Weibull shape parameter

Rule (NOTATION.md Rule 4). N is always the total sample size across randomization paths, never a per-path count. Table 1 gives the path count P for each design; the per-path allocation is $\lfloor N/P \rfloor$ with the remainder spread one per path.

Table 1: Trial designs and their randomization-path counts.

Design	P	Paths
OL (open-label)	1	on drug throughout
CO (crossover)	2	drug-then-placebo, placebo-then-drug
OL+BDC (blinded discontinuation)	2	discontinuation at either of two visits
Hybrid	4	two discontinuation points $\times$ two crossover orders

This distinction is not cosmetic: NOTATION.md documents a 2026-08-08 compendium-wide correction in which several drivers had read `modelparam$N` as a per-path count rather than a total. Re-matching every design to a common total $N = 70$ moved Paper 01’s Hybrid-design power from a ceiling of ≈ 0.99 down to 0.84 (tripling its measured carryover loss) and withdrew Paper 09’s second headline conclusion, that its most powerful design was also its most dropout-robust — the earlier near-immunity turned out to be a ceiling effect at the wrong N . This is the compendium’s clearest demonstration that an unmatched design comparison does not merely inflate one arm; it can manufacture an apparent robustness finding out of nothing.

2.2 Outcome and latent components

Symbol	Meaning
Y_{it}	observed symptom score, the modelled outcome (code column Sx)
BL_i	participant-specific baseline level
BR_{it}	biological (pharmacological) response component
PB_{it}	placebo-belief response component
TV_{it}	time-variant natural-history component
ε_{it}	observation-level residual
u_i, b_i	participant random intercept

2.3 Treatment exposure and carryover

Symbol	Meaning
D_{it}	binary drug state: 1 on drug, 0 off drug
$D_{bc,it}$	continuous exposure-decayed drug indicator (code <code>Dbc</code>)
t_{sd}	time since discontinuation (code <code>tsd</code>)
t_{od}	cumulative time on drug (code <code>tod</code>)
$t_{1/2}$	carryover half-life (code <code>carryover_t1half</code>)
λ	carryover decay rate, $\lambda = \ln 2/t_{1/2}$ (code <code>lambda_cor</code>)

Units. $t_{1/2}$ is quoted in *weeks* in the interaction-focused manuscripts (canonical grid $\{0, 0.5, 1.0\}$) and in *days* in the main-effect and test-procedure manuscripts, on a pharmacokinetic scale with a 3-day baseline (1 week = 7 days).

2.4 Biomarker, moderation, and estimand

Symbol	Meaning
B_i	participant-level pre-treatment biomarker (code <code>bm</code>)
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$ (code <code>bm_z</code>)
c_{bm}	covariance-moderation parameter: on-drug $\text{Cor}(B_i, BR_{it})$
β_{bm}	mean-moderation parameter: multiplier scaling the treatment effect by b_i
$c_{bm,A}, c_{bm,B}$	independent mean-channel and covariance-channel weights (dual-channel)
$\beta_{bm:D}$	the target estimand, the biomarker-by-treatment interaction (code <code>bm:Dbc</code>)
π	power, $\Pr(p < \alpha)$; $\alpha = 0.05$ throughout

Rule (NOTATION.md Rule 2). c_{bm} is a correlation; β_{bm} is a multiplier. They are calibrated to a common numeric scale — the reference value 0.45 denotes matched moderation strength under either architecture, because the mean-channel shift is applied as $\beta_{bm} b_i \sigma_{BR}$ on the standardized biomarker — but the two symbols are not interchangeable, and a paper using both architectures at a matched value must say so explicitly.

3 The response trajectory: modified-Gompertz progression

Every latent factor in the outcome model — BR , PB , and TV — is generated as a trajectory in t (or in t_{od} , cumulative time on drug) that rises from zero toward an asymptote. The default functional form throughout the package is a *modified Gompertz curve*; Paper 07 exists specifically to ask how much of the compendium’s power and bias findings survive if that functional-form choice is wrong.

3.1 The modified Gompertz curve

Paper 07’s canonical statement, with a vertical offset so the curve passes through the origin:

$$y(t) = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)), \quad y(0) = 0 \quad (1)$$

with maxr the asymptotic maximum response, rate governing the approach speed, and disp the curvature near $t = 0$; the inflection occurs at $t^* = \log(\text{disp})/\text{rate}$.

Paper 06’s equivalent statement makes the offset explicit and gives it as a normalized curve on $[0, 1]$ before scaling:

$$C(t) = m_C \exp(-d_C \exp(-r_C t)) - m_C \exp(-d_C), \quad (2)$$

for a generic component $C \in \{BR, PB, TV\}$, where m_C is the scale/asymptote parameter, d_C the displacement, and r_C the onset rate; the offset $m_C \exp(-d_C)$ is precisely the value the unshifted curve would take at $t = 0$, subtracted so $C(0) = 0$. Equations (1) and (2) are the same curve; NOTATION.md’s own shorthand is $y = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t))$ with a vertical offset adjustment, matching both.

Paper 05’s variant, folded into a normalized $[0, 1]$ scale, is algebraically the same curve written as a min-max normalization:

$$y(t) = \text{maxr} \cdot \frac{\exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)) - \exp(-\text{disp})}{1 - \exp(-\text{disp})}. \quad (3)$$

3.2 Alternative response-curve families

Paper 07 evaluates three alternatives, each calibrated so its asymptotic maximum and its response level at a single anchor time ($t = 5$ weeks) match the Gompertz reference:

$$\text{Logistic:} \quad y(t) = \frac{\text{maxr}}{1 + \exp(-\text{rate} \cdot (t - t_0))} - \text{offset} \quad (4)$$

$$\text{Hyperbolic tangent:} \quad y(t) = \text{maxr} \cdot \frac{\tanh(\text{rate} \cdot (t - t_0)) + 1}{2} - \text{offset} \quad (5)$$

$$\text{Piecewise-linear breakpoint:} \quad y(t) = \text{maxr} \cdot \min\left(1, \frac{t}{t_{\text{bp}}}\right) + \text{post_slope} \cdot \max(0, t - t_{\text{bp}}) \quad (6)$$

The logistic and hyperbolic-tangent curves are both symmetric around an inflection at t_0 ; the piecewise-linear form is the only non-differentiable member of the family, with a breakpoint at t_{bp} . NOTATION.md additionally names a **piecewise-linear-breakpoint** family among the package’s built-in `br_family` options alongside logistic and hyperbolic tangent; Paper 07 confirms the hyperbolic-tangent form is formally a rescaled logistic, not an independent shape family.

3.3 How trajectory shape interacts with carryover and the two architectures

Under the *mean-moderation* architecture the biomarker-dependent component of the treatment effect rides the trajectory itself:

$$Y_{it} = \mu_0 + \beta_1 D_{it} (1 + \beta_{bm} B_i) + \varepsilon_{it}, \quad (7)$$

so a family with a slower early rise front-loads less of the interaction signal onto the earliest on-drug timepoints; Paper 07 normalizes the per-timepoint moderation so its mean on-drug magnitude equals σ_{BR} , holding the marginal interaction effect size fixed while letting the temporal profile vary by family shape.

Under the *covariance-moderation* architecture (§5), by contrast, the interaction lives in

$$\frac{\partial \text{Cor}(BM, BR)}{\partial D_{bc}}, \quad (8)$$

which is independent of the trajectory family BR follows: the BM–BR covariance block is constructed directly, not derived from the mean curve, so trajectory-shape misspecification cannot move the covariance-channel interaction test at all. This is the structural reason Paper 07’s 16-cell factorial (four families $\times$ two architectures $\times$ two biomarker-effect levels, $N = 70$, Hybrid OL+BDC design, $t_{1/2} = 0$ to isolate the trajectory effect from carryover) finds power differences of only a few percentage points across families under both architectures: the mean channel is only mildly shape-sensitive, and the covariance channel is not shape-sensitive at all

by construction. Paper 07 is explicit that this 16-cell grid deliberately fixes $t_{1/2} = 0$; at $t_{1/2} > 0$ the **bm:Dbc**-mediated information channel would be eroded differently by different trajectory shapes, because the off-drug exponential decay interacts with each family’s own decay profile — a combination the present submission does not evaluate.

4 Trial designs, treatment exposure, and carryover

4.1 Simple versus complex designs

The compendium compares four designs of increasing complexity, defined by their randomization-path structure (Table 1) rather than by a closed-form exposure-sequence equation — no manuscript in the compendium formalizes D_{it} as an explicit function of path or sequence index; the designs are specified operationally, as visit-by-visit exposure schedules, and D_{it} is simply read off that schedule.

OL (open-label). Unblinded titration; no off-drug contrast, so it supplies no within-subject interaction information.

CO (crossover). Two-period within-subject crossover, blinded throughout.

OL+BDC (blinded discontinuation). Open-label titration followed by a blinded switch to placebo.

Hybrid. The Hendrickson (2020) reference design: open-label titration, blinded discontinuation, then a brief crossover; four randomization paths.

Design *complexity*, in this compendium, is therefore a statement about P and about how many within-subject on/off contrasts a design’s path structure supplies — not about a formal information functional of P . Paper 05, whose remit is exactly this comparison, contains no equation relating P or switch count to the power of the interaction estimand $\beta_{bm:D}$; its own stated target is in fact the *main* drug effect, not the interaction (that comparison is explicitly deferred to the companion architecture manuscript, Paper 01). The one quantitative link Paper 05 does supply is between serial correlation and the effective information in a *between*-subject contrast — relevant to why the parallel-group RCT loses power under AR(1) autocorrelation while the within-subject N-of-1 designs do not:

$$\text{effective-}N \text{ shrinkage factor} = \frac{1}{1 + (k - 1)\rho}, \quad (9)$$

where k is the number of repeated observations contributing to a between-subject mean and ρ is the AR(1) correlation. Because the N-of-1 designs’ primary contrast is within-subject, this shrinkage does not act on their information the way it acts on a parallel-arm comparator’s.

4.2 Treatment exposure: binary versus continuous

Rule (NOTATION.md Rule 1). D_{it} is binary; $D_{bc,it}$ is continuous. These are different objects and must never be substituted for one another:

$$D_{bc,it} = \begin{cases} 1 & D_{it} = 1 \text{ (on drug)} \\ \exp(-\lambda t_{sd,it}) & D_{it} = 0 \text{ (off drug)}, \end{cases} \quad (10)$$

with $t_{sd,it}$ the time since discontinuation. In the limit of an assumed zero half-life, the continuous predictor collapses exactly onto the binary one, $D_{bc,it} \rightarrow D_{it}$ as $\hat{t}_{1/2} \rightarrow 0$ (Paper 02) — the sense in which the compendium’s “Unadjusted” analysis specification is the boundary member of the continuous- exposure family.

4.3 Carryover decay families

Four decay forms are implemented, all normalized so $\phi(t_{1/2}) = 0.5$:

$$\text{Exponential: } \phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}} \quad (11)$$

$$\text{Linear: } \phi_{\text{lin}}(t_{sd}) = \max\left(0, 1 - \frac{t_{sd}}{2 t_{1/2}}\right) \quad (12)$$

$$\text{Weibull: } \phi_{\text{weib}}(t_{sd}) = \exp(-(\lambda_w t_{sd})^k), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}} \quad (13)$$

$$\text{Power: } \phi_{\text{pow}}(t_{sd}) = \max\left(0, \left(1 - \frac{t_{sd}}{3 t_{1/2}}\right)^p\right) \quad (14)$$

The Weibull family recovers the exponential exactly at shape $k = 1$; a value $k < 1$ produces a heavier elimination tail than exponential, and $k > 1$ a lighter one. Paper 02, whose remit is this decay family, tests $k \in \{0.25, 0.5, 2.0, 4.0\}$ against the exponential default and finds the analysis-model conclusions materially unchanged — the compendium’s carryover-specification results are not an artifact of assuming the exponential form. Paper 05 additionally implements a carryover *recursion* for the biological-response mean during off-drug periods,

$$\mu[t] = \text{base}[t] + \mu[t - 1] \cdot \phi(t_{sd}), \quad (15)$$

so that off-drug decay compounds across consecutive off-drug timepoints rather than resetting at each one.

4.4 Covariance moderation with carryover

Under the covariance-moderation architecture (§5), carryover attenuates the biomarker–response correlation itself, not just the mean:

$$\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd}), \quad (16)$$

with ϕ any of Equations (11)–(14) (Paper 02); in the exponential-only presentation of Paper 01 this is written $\text{Cor}(B_i, BR_{it}) = c_{bm}$ on drug and $c_{bm} \cdot e^{-\lambda t_{sd}}$ off drug. The conditional-mean interpretation of this construction, useful for connecting it to the mean-moderation architecture, is derived in Paper 01 as

$$E[BR_{it} \mid B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} (b - \mu_B), \quad (17)$$

with the off-drug analogue carrying the same decay factor $c_{bm} \cdot \phi(t_{sd})$ in place of c_{bm} .

4.5 Analysis specifications for carryover: E1–E9

Paper 02 fixes the analysis model’s carryover treatment across nine specifications, all fit as variants of

$$\text{nlme::lme}(\text{Sx} \sim \text{bm} * \text{X} + \text{t}, \text{random} = \sim 1 \mid \text{ptID}, \text{correlation} = \text{corCAR1}(\text{form} = \sim \text{t} \mid \text{ptID})) \quad (18)$$

differing only in the construction of the exposure predictor X and, for the CR-variant rows, the standard-error estimator (§10.3). Table 2 reproduces the full specification table.

E3, the lag-adjusted (Jones–Kenward) specification. The nuisance indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ marks off-drug timepoints that immediately follow an on-drug one; it enters as a main effect only, so the interaction actually tested remains $bm:D_{it}$, not $bm:L_{it}$.

Table 2: The nine carryover analysis specifications (Paper 02).

Code	Name	X / model	Target
E1	Unadjusted	$X = D_{it}$ (binary), no carryover term	$bm:D_{it}$
E3	Lag-adjusted	$X = D_{it}$, plus nuisance $L_{it} = D_{i,t-1}(1 - D_{it})$	$bm:D_{it}$
E2	Exposure-weighted	$X = D_{bc,it}$	$bm:D_{bc,it}$
E7	AIC-selected $\hat{t}_{1/2}$	E2 form, $\hat{t}_{1/2}$ chosen by AIC over $\{0.25, 0.5, 1.0, 2.0\}$ wk	$bm:D_{bc,it}$
E9	Paired-difference	two-stage OLS on per-patient on/off means	biomarker slope
E1cr2–E7cr2	CR2 variants	E1/E3/E2/E7 formula, CR2 cluster-robust SE	(as above)

E7, the AIC-selected half-life. With $M(t_{1/2})$ the E2 fit at a candidate half-life,

$$\hat{t}_{1/2} = \arg \min_{t_{1/2} \in \mathcal{T}} \text{AIC}(M(t_{1/2})), \quad \text{AIC}(M) = -2 \log L(M) + 2p, \quad \mathcal{T} = \{0.25, 0.5, 1.0, 2.0\}, \quad (19)$$

with the reported coefficient, SE, and p -value read from $M(\hat{t}_{1/2})$ alone — a post-selection-inference risk Paper 02 notes explicitly (Berk et al., 2013): treating $\hat{t}_{1/2}$ as fixed in advance understates the true sampling variability of the reported interaction estimate.

E9, the paired-difference specification. A correlation- structure-agnostic two-stage estimator. For patient i with $n_i^{\text{on}}, n_i^{\text{off}} > 0$,

$$\bar{S}_{x,i}^{\text{on}} = \frac{1}{n_i^{\text{on}}} \sum_{t: D_{it}=1} S_{x,it}, \quad \bar{S}_{x,i}^{\text{off}} = \frac{1}{n_i^{\text{off}}} \sum_{t: D_{it}=0} S_{x,it}, \quad \Delta_i = \bar{S}_{x,i}^{\text{on}} - \bar{S}_{x,i}^{\text{off}}, \quad (20)$$

followed by a second-stage OLS regression on the biomarker,

$$\Delta_i = \beta_0 + \beta_1 bm_i + \varepsilon_i, \quad i = 1, \dots, N', \quad (21)$$

with the interaction test on β_1 and $N' \leq N$ patients contributing (those without both on- and off-drug observations are dropped).

An earlier, retired pair of specifications, E5 (adding $bm:L_{it}$) and E6 (adding $bm:t_{sd}$ as a direct carryover-by-biomarker interaction), underperformed the Unadjusted specification at every cell tested and were replaced by E7/E9; they are recorded here as historical context, not part of the active nine-specification set.

5 Two architectures for a biomarker-treatment interaction

Paper 01 is the compendium’s central methodological result: the same target estimand, $\beta_{bm:D}$, can be encoded into the data- generating process in two structurally different ways, and the two encodings are *not* numerically interchangeable once carryover is present.

5.1 Mean moderation (Architecture A)

The biomarker scales the treatment effect directly, in the outcome’s conditional mean:

$$Y_{it} = \mu_0 + \beta_1 D_{it} (1 + \beta_{bm} B_i) + \varepsilon_{it}. \quad (22)$$

This is algebraically equivalent to the standard additive effect-modification form of Kent et al.,

$$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \varepsilon_{it}, \quad \beta_3 = \beta_1 \beta_{bm}, \quad (23)$$

identifying the interaction coefficient β_3 with the product of the base treatment effect and the moderation multiplier. The mean- difference-between-participants identity that follows,

$$E[Y | b_1, D] - E[Y | b_2, D] = \beta_1 \beta_{bm} (b_1 - b_2) D, \quad (24)$$

shows that under mean moderation, carryover (which shrinks D 's continuous analogue D_{bc} toward zero) shrinks the moderation contrast *proportionally* for every biomarker level — the mechanism is not selectively erased for any particular pair of participants.

5.2 Covariance moderation (Architecture B)

The biomarker and the biological-response component are instead drawn jointly from a bivariate normal whose correlation, not whose mean, is treatment-state dependent:

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN} \left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right), \quad \text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & D_{it} = 1 \\ c_{bm} \cdot e^{-\lambda t_{sd}} & D_{it} = 0. \end{cases} \quad (25)$$

The corresponding conditional-mean expressions on and off drug are Equations (17) and its carryover-decayed analogue,

$$E[BR \mid b, D = 0, t_{sd}] = \mu_{BR, \text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} (b - \mu_B). \quad (26)$$

Here the biomarker-dependent part of the conditional mean decays *exponentially* with time off drug, not proportionally: as t_{sd} grows, the differential signal between biomarker strata is eliminated entirely, in a way Equation (24)'s mean-moderation contrast never is. This structural asymmetry is the reason the two architectures diverge sharply once carryover enters: in Paper 01's matched- $N = 70$ headline result, carryover power loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ weeks in the Hybrid design is 1.4–6.9 percentage points under mean moderation but 5.1–15.4 points under covariance moderation, with a difference-of-differences $z = 2.80$ (Hybrid) and $z = 7.64$ (OL+BDC).

5.3 The common analysis model

Both architectures are fit identically:

$$\text{Sx} \sim \text{bm} + \text{t} + \text{Dbc} + \text{bm:Dbc} \quad (27)$$

via `nlme::lme` with `corCAR1(form = ~t|ptID)` residual correlation and a participant random intercept — the shared inferential machinery of §10 does not know, and cannot distinguish, which architecture generated the data it is fitting.

[Gap closed.] Paper 01's Methods now state the closed-form reference value explicitly, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$, alongside its bias performance measure, with a citation to the same calibration used in Paper 02 (Equation (56) below) and Paper 06 (Equation (43)). This is consistent with Paper 01's reported mean estimates of $\hat{\beta}_{bm:D_{bc}} \approx -0.23$ to -0.28 ; the manuscript itself does not evaluate σ_{BR} numerically, so the identity is stated symbolically there, exactly as in the two companion papers.

6 A latent-variable reading of covariance moderation

Paper 03 asks what mechanistic story a fitted covariance-moderation model could be a moment-matched summary of, if the true underlying process is a discrete responder/non-responder dichotomy rather than a continuously varying correlation.

6.1 The faithful latent-class data-generating process

If a candidate predictive biomarker indexes an unobserved responder/non-responder class,

$$Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \quad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc, it}), \quad (28)$$

with Z_i the unobserved class label and $\pi(B_i) = \Pr(Z_i = 1 \mid B_i)$ the class-membership probability, now given an explicit closed form in Paper 03’s Notation section,

$$\pi(B_i) = \text{expit}(\psi_0 + \psi_g b_i), \quad b_i = \frac{B_i - \bar{B}}{s_B}, \quad (29)$$

with ψ_g the **gating slope** (called **gs** in the code, and now given the symbol ψ_g in the manuscript rather than left unnamed) and ψ_0 fixed so $\pi(0) = 0.5$ throughout. f_0, f_1 are class-conditional response distributions differing in drug-response parameters. Within-class temporal structure follows the same modified-Gompertz form of §3, with class-specific parameters substituted.

[Notation caveat, resolved.] Paper 03 previously referred to gating-function steepness only informally, as **gs** in prose and code, with no manuscript symbol attached and no closed form for $\pi(B_i)$ ever written out — a gap this compendium originally flagged because a symbol γ *does* appear elsewhere in the compendium (Paper 06, §8) as the unrelated $BR \times PB$ subadditivity coefficient, inviting confusion between the two. Paper 03 now introduces ψ_g and Equation (29) explicitly, and its Notation section states in so many words that ψ_g is not to be confused with Paper 06’s γ . NOTATION.md has been updated to match: γ is now documented compendium-wide as Paper 06’s coefficient only, and ψ_g is added as its own entry for Paper 03’s gating slope.

6.2 The MVN approximation to mixture-induced moderation

Under within-class independence between B_i and BR_{it} , a two- component mixture with class-dependent means and common within-class covariance produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single MVN with $\text{Cor}(B, BR) = c_{bm}$, where

$$c_{bm}^2 = \underbrace{\frac{\pi(1-\pi)(\mu_B^1 - \mu_B^0)^2}{\text{Var}(B)}}_{\text{between-class fraction in } B} \cdot \underbrace{\frac{\pi(1-\pi)(\mu_{BR}^1 - \mu_{BR}^0)^2}{\text{Var}(BR)}}_{\text{between-class fraction in } BR}. \quad (30)$$

That is, c_{bm}^2 is the *product* of the two components’ between-class variance fractions, so the marginal correlation an MVN can reproduce is bounded above by the geometric mean of the two intra-class correlations. Equivalently: Paper 01’s Architecture B is, under this reading, the leading-order MVN signature of a latent-class mixture whose class-conditional means separate on drug and whose separation is eroded by carryover off drug — not necessarily a directly generative covariance mechanism in its own right. This reinterpretation gives covariance moderation a mechanistic story (discrete class membership) in addition to its literal one (a continuously specified correlation), and it identifies the conditions under which the approximation breaks down: strongly bimodal responses, a step-function (rather than smooth) $\pi(B_i)$, or class-varying (rather than common) within-class covariance.

6.3 Estimation and its two competing test statistics

The class-*unaware* analysis is the compendium’s common model, Equation (27). The class-*aware* analysis fits a two-class growth-mixture model (`1cmm:hlme`) and extracts two candidate signals: a Wald test on the biomarker’s coefficient in the class-gating equation, and a within-class Wald test on $bm:D_{bc}$. A model-comparison gate is applied before either is trusted:

$$\text{prefer } ng = 2 \text{ over } ng = 1 \iff \Delta\text{BIC} > 0 \text{ (or, more conservatively, } > 6), \quad (31)$$

following Raftery (1995), with the gating test interpreted only conditional on that selection. In Paper 03’s 240-replicate pilot at $N = 70$, the unconditional class-gating Wald test is markedly

anti-conservative under the null (Type I error 0.197, 95% CI [0.145, 0.250]), while the BIC-conditional version is under-powered relative to the class-unaware baseline (power 0.025 at gating slope 1.5 versus 0.542 for the ordinary *bm:Dbc* Wald test on the same cell) — the class-aware machinery, in this pilot, buys correct calibration at a substantial power cost, and is not obviously the better default test.

7 The dual-channel architecture

Paper 11 activates the mean and covariance channels of §5 simultaneously, with independent weights $c_{bm,A}$ (mean channel) and $c_{bm,B}$ (covariance channel).

7.1 Construction

The covariance channel is built into the MVN sigma matrix exactly as in Equation (25), with c_{bm} replaced by $c_{bm,B}$. The mean channel is then applied as a post-draw additive shift at on-drug timepoints,

$$\Delta BR_{it} = c_{bm,A} \cdot z_{B,i} \cdot \sigma_{BR} \cdot \mathbf{1}[\text{on-drug}], \quad z_{B,i} = \frac{B_i - \mu_B}{\sigma_B}. \quad (32)$$

7.2 The channels add on the covariance scale

Each channel’s on-drug contribution to $\text{Cov}(B, BR)$ can be computed separately:

$$\text{Covariance channel alone:} \quad \text{Cov}(B, BR \mid \text{on-drug}) = c_{bm,B} \sigma_B \sigma_{BR} \quad (33)$$

$$\text{Mean channel alone:} \quad \text{Cov}(B, \Delta BR) = c_{bm,A} \sigma_B \sigma_{BR} \quad (34)$$

and, because the combined-architecture on-drug response is the sum of the correlated draw and the mean shift, the two contributions add:

$$\text{Cov}(B, BR_C \mid \text{on-drug}) = (c_{bm,A} + c_{bm,B}) \sigma_B \sigma_{BR}. \quad (35)$$

To first order the interaction test’s noncentrality is likewise additive across channels, $\eta_C \approx \eta_A + \eta_B$, and Paper 11 reports empirical *super*-additivity of power itself relative to the natural probability-scale benchmark: for rejection probability $\pi(\eta)$ as a function of noncentrality,

$$\pi(\eta_A + \eta_B) - \pi(0) > [\pi(\eta_A) - \pi(0)] + [\pi(\eta_B) - \pi(0)]. \quad (36)$$

At $(c_{bm,A}, c_{bm,B}) = (0.22, 0.22)$ in the Hybrid design, at the matched total $N = 70$, for example, observed power is 0.577 against an additive benchmark $p(0.22, 0) + p(0, 0.22) - p(0, 0) = 0.214 + 0.138 - 0.040 = 0.312$, an excess of 0.265 — direct confirmation of Equation (36).

7.3 Boundary reductions and an identifiability consequence, now stated

Setting either weight to zero recovers the corresponding pure architecture exactly, by direct substitution into Equations (32) and (25): $c_{bm,B} = 0$ recovers mean moderation; $c_{bm,A} = 0$ recovers covariance moderation. Paper 11 treats these boundaries as internal validation gates — a correctly implemented combined driver must reproduce the companion manuscripts’ pure-architecture power values there, run under matched design, sample-size, and calibration settings. Equation (35) shows that the combined on-drug covariance — and hence the fitted *bm:Dbc* coefficient — depends on $c_{bm,A}$ and $c_{bm,B}$ only through their *sum*, so a single fitted interaction coefficient cannot separately recover the two channel weights; this non-identifiability, previously left unstated in Paper 11’s §2.2 (“Boundary reductions and identifiability”), is now given

as an explicit consequence in the manuscript’s §3, immediately after the additive-covariance derivation.

[A second gap, investigated rather than left flagged.] Paper 11’s driver (`analysis/scripts/dgp-combined/01-run-combined-factorial.R`) already allocates N as a total, via the same `generate_data_multi_path()` helper the companion paper uses, and its stored output already includes the matched $N = 70$ cells — so the sample-size-convention explanation an earlier version of §4.2 gave for the boundary mismatch does not hold up. Reading the $N = 70$, E2-specification boundary cells directly from that stored output and comparing them to Paper 01’s $N = 70$ baselines shows the two still do *not* agree: at $t_{1/2} = 0$ the mean-moderation boundary reads 0.595/0.738/0.612 (CO/Hybrid/OL+BDC) against Paper 01’s 0.739/0.842/0.751, a gap of 10–19 percentage points, far beyond the ≈ 0.013 – 0.016 MCSE at these cell sizes. The manuscript’s §4.2 has been rewritten to report this finding rather than the superseded N -convention explanation, and now names the likely locus of the mismatch (a calibration difference in `resp_param`, `baseline_param`, or the nuisance correlations `c.cf1t/c.cfct/rho` between the two drivers) as an open item for a follow-up revision. Paper 11’s own 3×3 channel-weight power table has likewise been recomputed on the correct $N = 70$ data already sitting in that same output file, in place of the earlier, uncorrected $N = 35$ table.

8 Partitioning the treatment effect

Paper 06 asks what a single-indicator (“lumped”) analysis of the treatment effect and its biomarker interaction actually estimates when the true generative process has three distinguishable components.

8.1 The outcome decomposition

$$Y_{it} = \text{BL}_i - [BR_{it} + PB_{it} + TV_{it}] + \varepsilon_{it}, \quad (37)$$

with each component following the modified-Gompertz form of Equation (2), and PB_{it} additionally scaled by an *expectancy multiplier* $\eta(\text{phase})$ that operationalizes belief-driven placebo response (Kirsch, 1985):

$$PB(t \mid \text{phase}) = \eta(\text{phase}) \cdot m_{PB} \exp(-d_{PB} \exp(-r_{PB}t)) + (\text{offset}), \quad \eta = \begin{cases} 1.0 & \text{open-label, on drug} \\ \approx 0.5 & \text{blinded discontinuation} \\ \in (0, 0.5) & \text{open-label placebo.} \end{cases} \quad (38)$$

Because η scales both the mean and SD of PB by the same factor, $\text{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$, only the *product* $\eta \cdot m_{PB}$ is identified from a two-phase (open-label + blinded) design; recovering m_{PB} itself requires an open-label-placebo phase or a direct expectancy measurement.

A generalized, non-additive form allows a $BR \times PB$ interaction:

$$Y_{it} = \text{BL}_i - [BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}] + \varepsilon_{it}, \quad (39)$$

with *subadditivity* of the two response mechanisms defined by the sign condition $\partial Y / \partial (BR \cdot PB) > 0$ on the reduction scale. Paper 06 explicitly distinguishes true $\gamma \neq 0$ subadditivity from an *apparent* subadditivity that a latent responder-class gating function $\pi(B)$ (§6) could generate through confounding, without any real BR – PB interaction being present.

8.2 What a one-component analysis actually estimates

Writing the within-participant symptom reduction as $R_{it} = \text{BL}_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$ and fitting the one-component model $R_{it} = \alpha + \beta_D D_{it} + g(t) + u_i + e_{it}$, Paper 06 derives:

$$\text{plim } \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV}, \quad (40)$$

where $b_{PB} = E[PB \mid \text{on}] - E[PB \mid \text{off}]$ and b_{TV} is the natural-history contribution that fails to cancel under the contrast used (zero for a clean on/off within-participant contrast; $E[TV]$ itself for a baseline-anchored pre-post analysis). Both terms are generally non-negative for an improving outcome, so the one-component estimate systematically *over-states* the pharmacological effect m_{BR} — and this is a property of the estimator’s probability limit, unaffected as $N \rightarrow \infty$, not a finite-sample artifact correctable by more data.

The same logic applied to the biomarker slope gives the compendium’s key attenuation/inflation identity:

$$\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB} \beta_{bm}^{PB} + w_{TV} \beta_{bm}^{TV}, \quad \beta_{bm}^C = \frac{\text{Cov}(m_C, B)}{\text{Var}(B)} \quad (C \in \{BR, PB, TV\}), \quad (41)$$

with $w_{PB}, w_{TV} \geq 0$ design-determined loadings. A biomarker correlated with placebo responsiveness or with the natural-history trajectory therefore registers a non-zero lumped slope even when its true pharmacological association β_{bm}^{BR} is exactly zero — and, because Equation (41) is an *identity*, no amount of additional sample size can separate a genuinely pharmacological biomarker signal from a confounded one whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$. Under $\gamma \neq 0$ sub-additivity (Equation (39)), a further first-order correction proportional to $\text{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the slope; the identity above is the leading-order behavior in the additive case.

The operative confound is a biomarker’s correlation with individual-level *variance* in placebo responsiveness, not with its *mean*:

$$\text{contamination driver} = c_{bm, PB} \cdot \sigma_{PB}, \quad \text{not } c_{bm, PB} \cdot m_{PB}. \quad (42)$$

Paper 06’s calibration target throughout its production studies is

$$\beta_{bm:D_{bc}} = -\frac{c_{bm} \sigma_{BR}}{\sigma_{bm}} = -2.25, \quad (43)$$

against which Study A (uncontaminated biomarker, $c_{bm, PB} = 0$) shows bias within one Monte Carlo standard error of zero at every sample size tested ($N \in \{35, 70, 200, 300\}$), confirming Equation (41) collapses to zero bias exactly when $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$; a contaminated-biomarker pilot at $c_{bm, PB} \in \{0, 0.15, 0.30, 0.45\}$ (constrained below 0.45 by positive-definiteness of the resulting covariance matrix) shows bias growing from -0.06 to $+0.51$ (about 30 MCSEs) as contamination increases, with the sign flip explained by an *attenuation- toward-zero* mechanism: high- $c_{bm, PB}$ participants’ elevated placebo response is partially absorbed by the biomarker main effect, leaving the interaction coefficient itself pulled toward zero rather than pushed away from it.

Critically, Paper 06 emphasizes throughout that this bias lives in the *location* of $\hat{\beta}_{bm:D_{bc}}$, not in the test’s power or rejection rate — a significance test of the lumped coefficient cannot, even in principle, detect this bias, because the bias does not move the point estimate toward the null.

8.3 Recovery, and its dependence on trial design

A phase-augmented analysis model is the smallest step toward component-aware inference:

$$\text{Sx} \sim \text{bm} + \text{t} + \text{Dbc} + \text{phase} + \text{Dbc:phase} + \text{bm:Dbc} + \text{bm:Dbc:phase} \quad (44)$$

with random intercept and `corCAR1` residual correlation, where `phase` (open-label / blinded / post-crossover) absorbs systematic expectancy differences and `Dbc:phase`, `bm:Dbc:phase` test whether the apparent drug effect and its moderation shrink under blinding. Whether this — or full component- matched Gompertz-basis fitting — actually recovers the true pharmacological slope β_{bm}^{BR} is shown to be *design-dependent*, not a property of decomposition per se: under the

compendium’s standard Hybrid design, belief-covariate decomposition, blinded-stratum contrasts, and full Gompertz-basis decomposition all fail to recover β_{bm}^{BR} , performing no better (and sometimes worse) than the lumped analysis, because D_{bc} and the expectancy multiplier η are positively correlated ($\approx +0.6$) under that design. A belief-decoupling *balanced-placebo* design — three open-label, six covert on-drug ($\eta = 0$), and six open-placebo ($\eta = 1$) timepoints — reduces that correlation to ≈ -0.66 and restores recovery: at matched N , decomposition-analysis bias stays within one MCSE of zero across $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ while the lumped estimator’s bias grows monotonically over the same grid.

9 Informative dropout

Paper 09 evaluates a censoring mechanism whose hazard depends on the participant’s own symptom trajectory, crossed with the four trial designs of §4.

9.1 The dropout hazard

Marginal per-timepoint censoring probability is the union of two independent components:

$$\Pr(\text{drop}_{it}) = 1 - (1 - p_{it}^{\text{flat}})(1 - p_{it}^{\text{bias}}), \quad (45)$$

with $p_{it}^{\text{flat}} \propto t$ depending on elapsed time alone (calibrated to a marginal dropout fraction of $\beta_0(1 - \beta_1)$) and $p_{it}^{\text{bias}} \propto t(\Delta_{Sx}(t) + 100)^{e_{b1}}$ depending on the symptom change $\Delta_{Sx}(t)$ since baseline (calibrated to $\beta_0\beta_1$), where β_0 is the overall dropout rate, β_1 the fraction attributable to the symptom- biased mechanism, and $e_{b1} = 2$ throughout. The additive shift of 100 inside the biased component keeps its convex argument positive across the realized range of symptom scores, so that large improvement lowers dropout probability while worsening (or a small change) raises it. Once a participant is censored at a timepoint, every later timepoint for that participant is also set missing (monotone/absorbing dropout).

9.2 Missingness classification

Because the flat component depends only on a fully observed design variable (elapsed time), it induces missingness that is ignorable for likelihood-based inference on its own. Because the biased component’s censoring indicator is a function of $\Delta_{Sx}(t)$ — the same quantity realized at, and thereby rendered missing by, that timepoint — Paper 09 argues the mechanism is MNAR for every dropout pattern with $\beta_1 > 0$: missingness of Y_{it} depends on the value of Y_{it} itself, the defining condition for missing-not-at- random data. No manuscript in the compendium gives a formal probability statement for MCAR/MAR/MNAR beyond this argument; the classification rests on the hazard construction of Equation (45) alone.

9.3 Interaction with design

The estimand is the same interaction test as elsewhere, $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$, fit post-dropout. Paper 09 documents a “happy accident” selection effect, described qualitatively rather than by a closed-form path-conditional formula: participants with unambiguous open-label response are the most readily resolved and most prone to drop out early, which enriches the later, randomized (crossover/discontinuation) blocks for participants whose open-label response was ambiguous — precisely the subgroup whose crossover contrasts are most informative for the interaction test. This selection effect is design-dependent: under the OL design, where D_{bc} is identically one throughout, there is no within- subject contrast for $bm:D_{bc}$ at all, so dropout there can only reduce power, never selectively concentrate information. A quantified, per-path decomposition of the happy-accident effect is explicitly scoped for a future production-grade rerun and is not delivered in the version inspected here; as with several other design

comparisons in this compendium (§2), an earlier mismatched- N run of this same comparison had manufactured an apparent robustness finding (the Hybrid design appearing dropout-immune) that a matched- $N = 70$ re-run showed was a ceiling effect rather than a real property of the design.

10 Inference: from a fitted coefficient to a calibrated test

10.1 A closed-form comparator: repeated-measures ANOVA

Paper 08’s principal analytic contribution, stated for comparison against the compendium’s simulation-based test procedures, is a full closed-form treatment of the classical split-plot RM-ANOVA under a dichotomized biomarker. The model:

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijkl}, \quad (46)$$

for biomarker stratum i , treatment phase j , subject l , and timepoint k , with $\pi_{l(i)} \sim N(0, \sigma_\pi^2)$ the random subject effect and $e_{ijkl} \sim N(0, \sigma_e^2)$ the within-subject error under compound-symmetric sphericity. The biomarker–treatment interaction $(\alpha\beta)_{ij}$ is the target; its test statistic is

$$F_{AB} = \frac{MS_{AB}}{MS_{B \times S(A)}} \sim F(df_{AB}, df_{B \times S(A)}) \text{ under } H_0, \quad (47)$$

which specializes, for $a = b = 2$ strata/phases and T within-phase timepoints, to a fully closed form. Writing the interaction contrast as $\hat{L} = (\bar{\Delta}_1 - \bar{\Delta}_2)$ with $\bar{\Delta}_i$ the stratum- i mean of each subject’s own phase-1-minus-phase-2 within-subject difference,

$$F_{AB} = \frac{nT \hat{L}^2}{4 \hat{\sigma}_e^2}, \quad t_{AB} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \quad t_{AB}^2 = F_{AB}, \quad df_1 = 1, \quad df_2 = 2(n-1), \quad (48)$$

recovering the algebraic equivalence of the 2×2 interaction F -test to a two-sample t -test on within-subject differences. Power follows the noncentral F distribution with noncentrality

$$\lambda = \frac{nT \Delta_0^2}{4 \sigma_e^2}, \quad 1 - \beta = \Pr[F_{AB} > F_{1, 2(n-1), 1-\alpha} \mid \lambda], \quad (49)$$

where Δ_0 denotes the true value of the contrast $\hat{L}$ under H_1 , an effect size on the raw outcome scale. Paper 08 originally called this quantity δ , which collided with the unrelated *standardized bias* δ of the general calibration framework in §10.2 below; the manuscript has been relabeled to Δ_0 , with an explicit note distinguishing it from Paper 10’s δ , so the symbol δ is now reserved compendium-wide for standardized bias alone. Paper 08 finds this closed form is the compendium’s only fully analytic power result; GEE, Mancl–DeRouen, and the LME/corCAR1 model it compares against are evaluated purely by simulation, with the sandwich/CR2 machinery of §10.3 providing the small-sample correction.

10.2 A general decomposition of realized Type I error

Paper 10, written in deliberately general Wald-test notation (applicable to any fixed-effect interaction coefficient, not only $\beta_{bm:D}$), decomposes an estimator into a standardized component:

$$\hat{\theta} = \mu + \sigma Z^*, \quad \mu = E(\hat{\theta}), \quad \sigma = \text{SD}(\hat{\theta}), \quad Z^* = \frac{\hat{\theta} - \mu}{\sigma}, \quad (50)$$

and rewrites the Wald statistic $T = \hat{\theta}/\widehat{\text{SE}}$ in terms of two dimensionless channels,

$$T \approx \frac{\mu + \sigma Z^*}{\kappa \sigma} = \frac{1}{\kappa} (Z^* + \delta), \quad \kappa = \frac{\widehat{\text{SE}}}{\sigma}, \quad \delta = \frac{\mu}{\sigma}. \quad (51)$$

Realized Type I error follows directly:

$$\alpha_{\text{real}} = \Pr(|T| > t_{\nu, 1-\alpha/2}) \approx \Pr(|Z^* + \delta| > \kappa t_{\nu, 1-\alpha/2}), \quad (52)$$

and exactly *four* channels can move it away from nominal: a standardized bias δ , a scale-distortion κ (model-based SE too large or small relative to the true sampling SD), a degrees-of-freedom channel ν , and the shape of Z^* 's distribution (departures from normality). Calibration is recovered exactly when $\delta = 0$, $\kappa = 1$, $\nu = \infty$, and $Z^* \sim N(0, 1)$. With the bias and reference-distribution channels inert, this reduces to a closed form in κ alone,

$$\alpha_{\text{real}} \approx 2\Phi(-\kappa t_{\nu, 1-\alpha/2}) \approx 0.05 - 0.23(\kappa - 1) \text{ near } \kappa = 1, \quad (53)$$

with elasticity $d \log \alpha_{\text{real}} / d \log \kappa \approx -4.6$ at $\kappa = 1$: a small overstatement of the model-based standard error produces a disproportionate collapse in realized Type I error, which is the compendium's general explanation for why an idealized OLS analysis that ignores the participant random effect is anti-conservative — its scale-distortion factor is approximately

$$\kappa_{\text{OLS}}^2 \approx \frac{\sigma_b^2 + \sigma_\varepsilon^2}{\sigma_\varepsilon^2}, \quad (54)$$

exceeding one by exactly the random-intercept variance it omits.

Paper 02 gives the compendium's operational estimator of κ directly on the interaction coefficient, using the notation $\hat{\beta}_{bm:X}$ for the fitted coefficient under carryover specification X :

$$\kappa = \frac{\widehat{SE}}{\text{SD}(\hat{\beta}_{bm:X})} = \frac{\frac{1}{n_{\text{sim}}} \sum_{r=1}^{n_{\text{sim}}} \widehat{SE}_r}{\sqrt{\frac{1}{n_{\text{sim}}-1} \sum_{r=1}^{n_{\text{sim}}} (\hat{\beta}_{bm:X,r} - \hat{\beta}_{bm:X})^2}}, \quad (55)$$

and the calibrated true value against which bias is computed throughout the carryover-specification manuscript,

$$\theta_{\text{true}} = -c_{bm} \sigma_{BR}. \quad (56)$$

10.3 Cluster-robust sandwich variance and the CR2 correction

The naive (Liang–Zeger) sandwich estimator, clustering on participant,

$$\widehat{\text{Var}}_{\text{sandwich}}(\hat{\beta}) = \left(\sum_i X_i' \hat{V}_i^{-1} X_i \right)^{-1} \left(\sum_i X_i' \hat{V}_i^{-1} \hat{e}_i \hat{e}_i' \hat{V}_i^{-1} X_i \right) \left(\sum_i X_i' \hat{V}_i^{-1} X_i \right)^{-1}, \quad (57)$$

is downward-biased at moderate cluster counts. Paper 10 gives the general CR family in the GLS-transformed metric,

$$\hat{V}_{\text{CR}} = (\tilde{X}' \tilde{X})^{-1} \left(\sum_i \tilde{X}_i' A_i \tilde{e}_i \tilde{e}_i' A_i \tilde{X}_i \right) (\tilde{X}' \tilde{X})^{-1}, \quad (58)$$

indexed by the adjustment matrix A_i : $A_i = I$ (CR0, unadjusted); $A_i = (I - H_{ii})^{-1}$ (the jackknife form, equal to the Mancl–DeRouen correction in the marginal/GEE setting); and

$$A_i = (I - H_{ii})^{-1/2} \quad (\text{CR2}), \quad (59)$$

the symmetric square root that renders $\hat{V}_{\text{CR}}$ exactly unbiased when the working covariance is correctly specified — the cluster generalization of the HC2 estimator of MacKinnon and White. Its Satterthwaite degrees of freedom are recovered by matching the first two moments of the quadratic form's eigenvalues λ_g ,

$$\nu = \frac{(\sum_g \lambda_g)^2}{\sum_g \lambda_g^2}. \quad (60)$$

Paper 02's null-cell diagnostics find κ in the range 1.05–1.10 (empirical Type I error 0.029–0.039 against nominal 0.05) for the model-based standard errors, and κ 1.07–1.26 more broadly across specifications — consistent, via Equation (53), with the mild conservatism the compendium's Wald tests generally exhibit rather than anti-conservatism.

11 The treatment main-effect track

Paper 04 targets a different estimand under a different, simpler DGP: the treatment *main* effect β_D , with **no biomarker moderation channel at all**, and a linear (not Gompertz) data-generating model.

11.1 Data-generating model

For the N-of-1 arm,

$$Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \beta_t t_{ij} + u_i + \varepsilon_{ij}, \quad u_i \sim N(0, \sigma_u^2), \quad (61)$$

with ε_i following a continuous-time AR(1) covariance $\text{Cov}(\varepsilon_j, \varepsilon_k) = \sigma_w^2 \rho^{|t_j - t_k|}$; carryover is applied as an additive exponential decay to the period immediately following a drug period, $\delta_c = \beta_D \exp\{-\ln(2) L/t_{1/2}\}$ with L the period length in days. For the comparator parallel-group RCT arm, drawing patient baselines from a shared pool,

$$Y_i = \mu_0 + u_i + \beta_D Z_i + \varepsilon_i, \quad u_i \sim N(\mu_0, \sigma_\tau^2), \quad Z_i \in \{0, 1\}. \quad (62)$$

The two arms share the same $N = 35$ patient pool (participant-count- matched, not observation-count-matched — flagged by the paper itself as a limitation), fit respectively by `nlme::lme(Sx ~ Dbc + t_wk, ..., correlation = corCAR1(...))` with Satterthwaite df, and by ANCOVA, `lm(endpoint ~ trt + baseline)`.

[Internal inconsistency, resolved.] Paper 04 previously stated $\sigma_w = 1.0$ in the AR(1) covariance definition of its DGP (§2.1) while separately citing $\sigma_w = 2.3$ as “moderate within-patient measurement noise” in its optimal-design-conditions discussion (§6), with the two values never reconciled in the source text. Inspection of the production driver (`analysis/scripts/treatment-main-effect/pro` constant `SIGMA_WITHIN <- 2.3`) confirms $\sigma_w = 2.3$ is the value the simulation actually uses; §2.1’s $\sigma_w = 1.0$ was the error, and has been corrected in the manuscript to match both §6 and the code.

11.2 Classical closed-form power theory (background, not the paper’s own simulation)

Paper 04’s literature-review section reproduces the closed-form power apparatus this compendium otherwise replaces with simulation, due to Senn (2018) and Zucker et al.:

$$d_{ij} = \delta_i + \varepsilon_{ij}, \quad \delta_i \sim N(\Delta, \tau^2), \quad \varepsilon_{ij} \sim N(0, 2\sigma^2) \quad (63)$$

$$\text{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n} \quad (\text{random-effects, patient-mean analysis}) \quad (64)$$

$$\lambda_{\text{RE}} = \frac{\Delta}{\sqrt{(\tau^2 + 2\sigma^2/k)/n}}, \quad df = n - 1 \quad (65)$$

$$\lambda_{\text{FE}} = \frac{\Delta}{\sqrt{2\sigma^2/(nk)}}, \quad df = n(k - 1) \quad (\text{fixed-effects, pools within-patient SS}) \quad (66)$$

for n patients and k paired treatment cycles each, with τ^2 the between-patient (treatment-by-patient interaction) variance and $2\sigma^2$ the within-patient, within-cycle variance of paired differences. The paper notes standard software assuming $n - 1$ denominator df underestimates power here whenever $k > 2$, because the fixed-effects analysis correctly pools within-patient degrees of freedom that the random-effects analysis discards. Zucker et al.’s equivalent precision formula,

$$\text{Var}(\hat{\Delta}) = \frac{\tau^2 + \sigma_w^2/k}{n}, \quad (67)$$

makes explicit that the marginal benefit of additional cycles k diminishes once $\sigma_w^2/k \ll \tau^2$: past that point, adding cycles to existing patients buys far less precision than adding new patients, because the between-patient variance floor dominates.

Paper 04’s own simulation (2,000 replicates per cell, effect-size grid $\beta_D \in \{0, -0.5, -1.0, -2.0\}$ nightmares/week, carryover half-life sweep $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days) does not invoke Equations (64)–(67) directly; it estimates power and Type I error empirically, at the MCSE rates given in §12 below.

12 Shared performance measures

Every manuscript in the compendium reports the same three Monte Carlo performance quantities, following the ADEMP framework of Morris, White, and Crowther (2019):

$$\hat{\pi} = \frac{1}{n_{\text{reps}}} \sum_{r=1}^{n_{\text{reps}}} \mathbf{1}[p_r < \alpha] \quad (\text{empirical power / Type I error}) \quad (68)$$

$$\text{MCSE}(\hat{\pi}) = \sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}} \quad (69)$$

$$\text{Bias} = \frac{1}{n_{\text{reps}}} \sum_r \hat{\theta}_r - \theta_{\text{true}}, \quad \text{MCSE}(\text{Bias}) = \frac{s_{\hat{\theta}}}{\sqrt{n_{\text{reps}}}} \quad (70)$$

$$\text{MCSE}(\widehat{SE}) = \frac{s_{\hat{\theta}}}{\sqrt{2(n_{\text{reps}} - 1)}} \quad (71)$$

with $s_{\hat{\theta}}$ the empirical standard deviation of the replicate-level point estimates. A minimum replicate count of $n_{\text{reps}} \geq 625$ is the compendium’s recurring target, derived from requiring $\text{MCSE}(\hat{\pi}) < 0.02$ at the worst-case $\hat{\pi} = 0.5$; most production runs use $n_{\text{reps}} \in \{500, 1000\}$ for power/bias cells and up to 5000 for null/Type-I cells, where the smaller true $\hat{\pi}$ near α tightens the MCSE requirement at fixed sample size.

For a difference-of-differences comparison across two conditions (e.g. the carryover-loss architecture contrast of §5), Paper 01 gives

$$\text{SE}(\Delta_B - \Delta_A) = \sqrt{\frac{\hat{\pi}_{B,0}(1 - \hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1 - \hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1 - \hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1 - \hat{\pi}_{A,1})}{n}}, \quad \Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}, \quad (72)$$

the standard error of the within-architecture carryover-loss gap, used throughout the compendium to test whether two designs’, architectures’, or specifications’ power differences are distinguishable from Monte Carlo noise.

Summary: which manuscript supplies which piece

Paper	Contribution to this compendium	Key equations
01	Mean- vs. covariance-moderation architectures; the central carryover-divergence result	(22)–(26)
02	Carryover decay families; nine analysis specifications E1–E9; κ calibration; sandwich/CR2	(11)–(21), (55)–(56)
03	Latent-class reading of covariance moderation; MVN moment-matching approximation	(28)–(31)

Paper	Contribution to this compendium	Key equations
04	Treatment main effect; simpler linear DGP; classical Senn/Zucker power theory	(61)–(67)
05	Simple vs. complex designs; effective- N shrinkage under AR(1); Gompertz and carryover-recursion restatements	(9), (3), (15)
06	Partitioning the treatment effect into $BR/PB/TV$; lumped- slope bias identity; expectancy scaling	(37)–(44)
07	Modified-Gompertz curve and alternative trajectory families; architecture-dependent shape sensitivity	(1)–(8)
08	Closed-form RM-ANOVA derivation; the compendium’s only fully analytic power result	(46)–(49)
09	Informative-dropout hazard; MNAR argument; design-dependent “happy accident” selection effect	(45)
10	General four-channel (δ, κ, ν , shape) decomposition of realized Type I error; CR family	(50)–(60)
11	Dual-channel architecture; additive covariance, super-additive power, unstated non-identifiability	(32)–(36)